PRISM | Tech Management, SRI-B

CLASH OF THE **CLAWS**

RVCE_QuadNova
“Beyond ATS. Beyond Keywords”

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Problem Statement

Why did we select the "Productivity Platforms" theme?

74% of companies admit to making "bad hires," often due to unverified claims. Keyword-based ATS filters reject 94% of resumes before human review, creating a massive loss of high-signal talent.

Who faces this problem?

Recruiters / Hiring Managers

- resume overload,
- manual verification effort,
- difficulty trusting resumes,
- and pressure to hire quickly.

Candidates / Students

- unfair ATS rejection,
- lack of feedback,
- inability to showcase practical skills,
- and confusion about what skills they lack.

What problem are you solving?

We are solving the problem of:
"Untrustworthy and inefficient hiring."
Traditional ATS systems depend heavily on resumes, which are self-declared and often exaggerated.

Current systems fail because:

- No real skill validation
- Ignores GitHub & portfolios
- Keyword-based candidate rejection
- No hiring transparency

This creates:

- bad hires,
- recruiter overload,
- unfair filtering,
- and poor candidate experience

Why is it important?

Hiring directly impacts productivity, innovation, and business growth. Traditional ATS systems often miss genuine talent, leading to poor hiring decisions, wasted resources, and inefficient recruitment processes.

Current Solution & Gaps

Solution Category	Representative Examples / Research	Capabilities Demonstrated	Critical Gaps Relative to Problem Statement
Commercial AI Hiring Platforms	Eightfold AI, Beamery, SeekOut	Deep skill inference from resumes and career histories; semantic search; talent graph construction; basic fit scores	Limited cross-source skill normalization (e.g., GitHub READMEs vs. LinkedIn); explanations lack counterfactual reasoning ("You scored 72% because missing cloud architecture reduces match by 18 points"); poor handling of non-linear career trajectories (career switchers, gig workers)
Academic Research Prototypes	GraphRank Pro+ 2, DISCO 9, TalentBERT (KDD 2022, expert materials)	Knowledge graph-based profiling; disentangled cognitive diagnosis for interpretable recommendations; BERT-based semantic alignment	Remain largely non-deployed; lack integration with heterogeneous data sources (portfolios, bootcamp certificates); no end-to-end validation linking explainable scores to hiring outcomes (time-to-fill, retention)
NLP-Powered Resume Analyzers	AI-Powered Resume Analysis using SpaCy 1, ResumeMatch 3, CredMatch 10	Accurate extraction of skills/experience from unstructured text; bias mitigation by ignoring demographic details; personalized resume enhancement suggestions	Focus primarily on resume-to-job description matching; do not incorporate external evidence of skill (e.g., code repositories, design portfolios); explanations are descriptive, not diagnostic
Skill Gap & Learning Systems	Skill Gap Analysis Using ML 8, AI Career Navigator 11	Identification of missing skills; recommendation of learning resources; career path prediction	Reactive rather than proactive in matching; not integrated into real-time job recommendation engines; lack dynamic updating as candidate skills evolve
Explainable AI (XAI) Frameworks	LEMON 6, EvalxNLP 12, Feature Responsiveness Scores 13	Model-agnostic explanation methods; benchmarking of post-hoc explainability; identification of "principal reasons" for adverse decisions	Not specifically tailored to recruitment domain; explanations often fail to translate technical feature attributions into actionable, human-understandable insights for candidates or recruiters

Your Solutions

What we built

We built Proof-Hire, an AI-powered talent intelligence platform where recruiters can create job openings and candidates can apply using their resumes. The system analyzes resumes, GitHub profiles, and portfolios to generate detailed candidate insights, fit scores, and skill-gap analysis.

Core Ideas Explanation

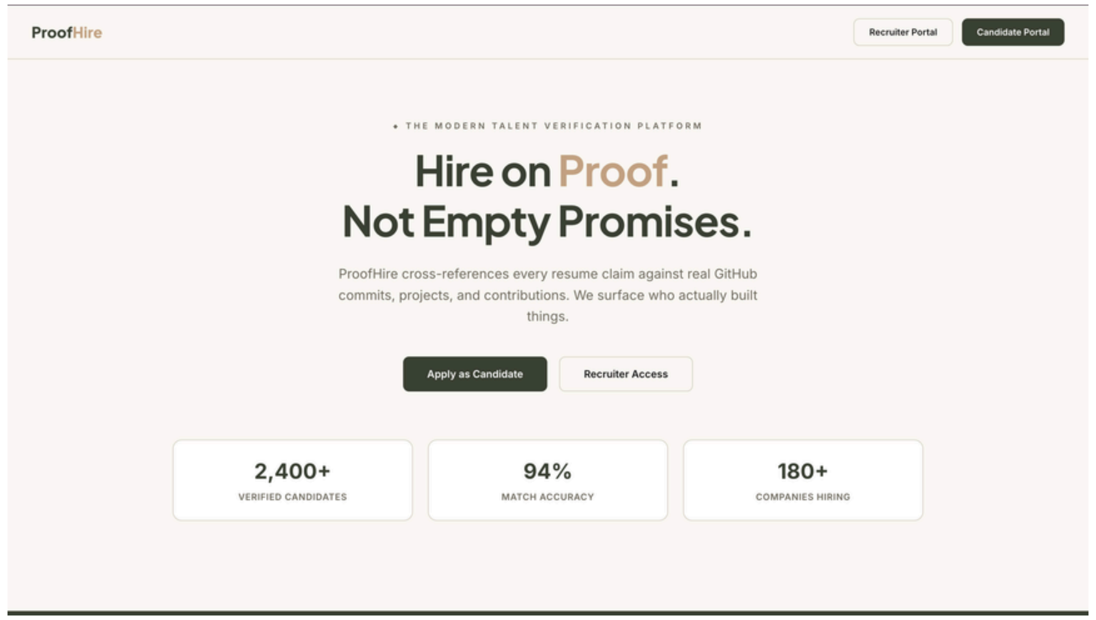
- Use proof-based hiring through GitHub and project analysis
- Provide explainable AI-driven candidate scoring
- Analyze resumes semantically instead of exact keyword matching
- Help recruiters make faster and smarter hiring decisions
- Validate real technical skills using repositories and activity
- Help candidates understand skill gaps and improve through personalized roadmaps

Key Features

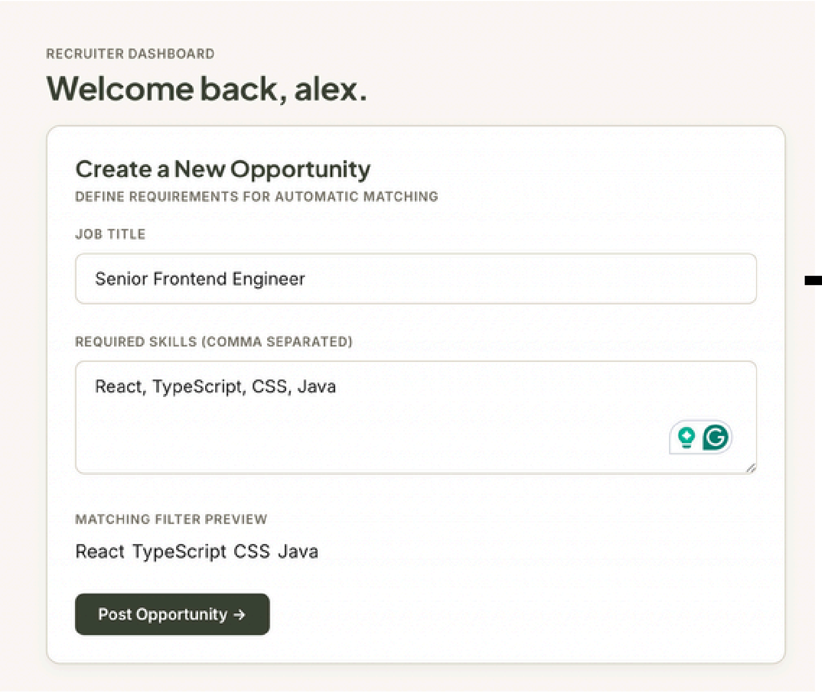
- Recruiter job creation with required skills
- Resume parsing and candidate summarization
- Automatic GitHub & LinkedIn link extraction
- GitHub analysis: repositories, activity, stars, languages used
- AI-powered semantic skill matching
- Candidate scoring based on skills, GitHub, and CGPA
- Interactive knowledge graph visualization
- Explainable hire/reject decision support
- Personalized roadmap for candidate improvement

Proof-Hire : Multimodal Talent Intelligent Platform

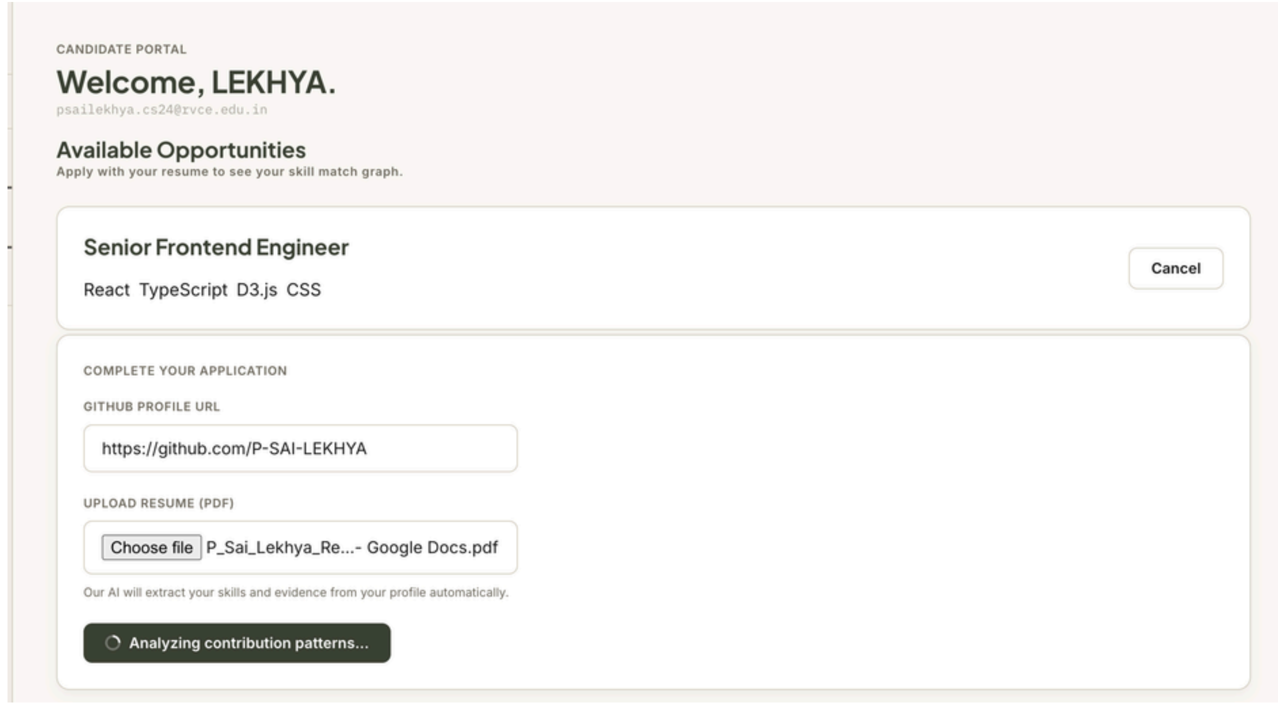
Demo/Product Walkthrough



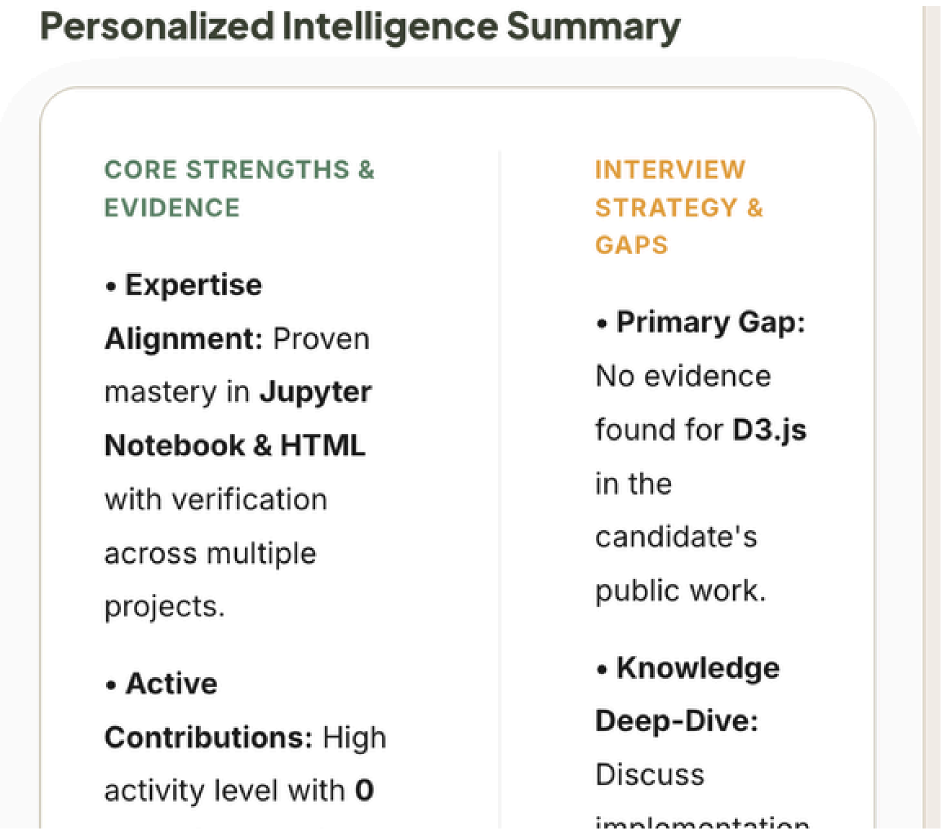
Recruiter Portal and Candidate Portal



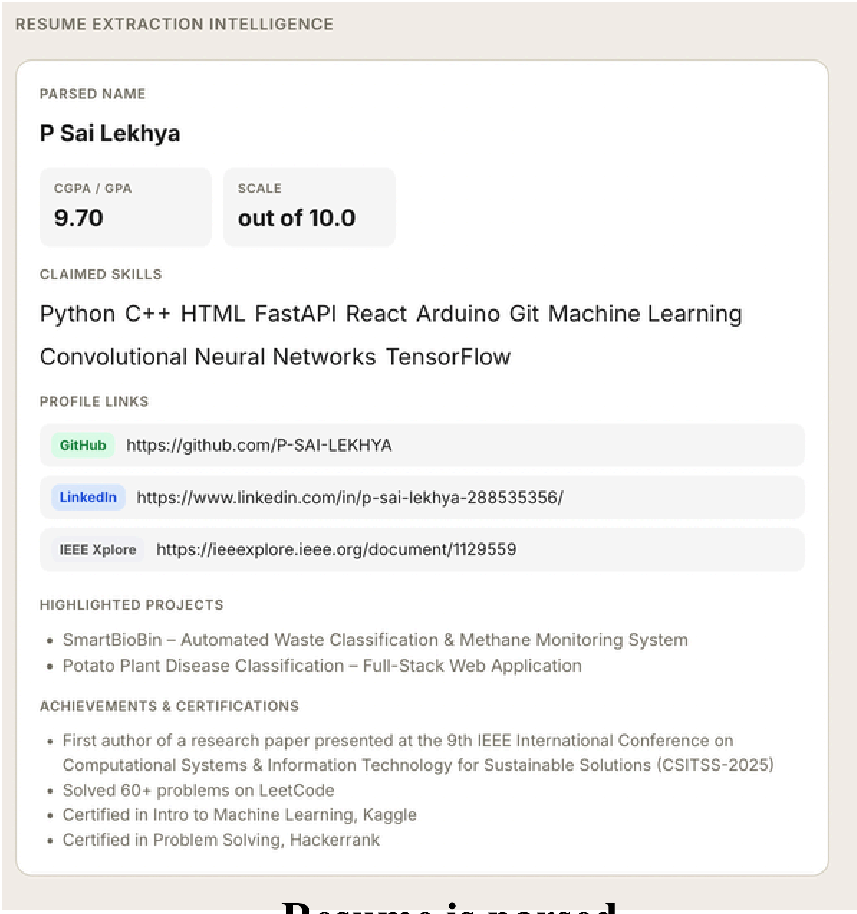
Recruiter creates Job opportunities



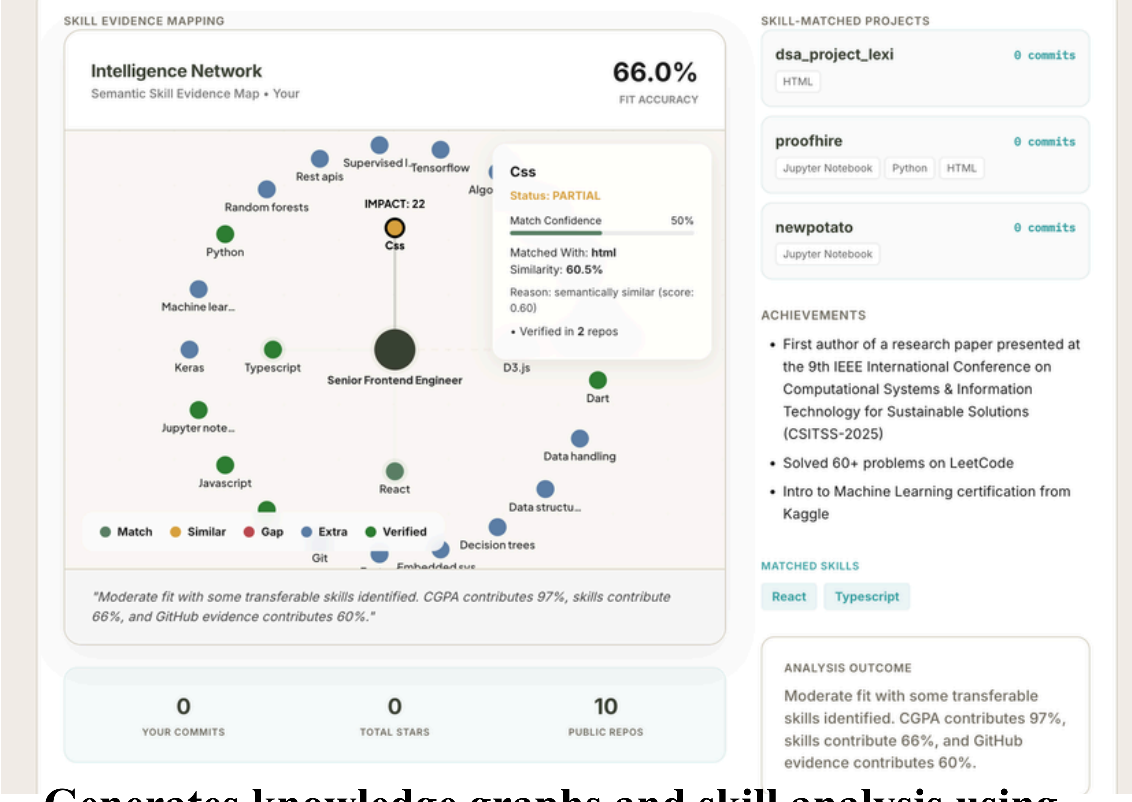
AI-powered resume parsing with automatic skill and profile link extraction.



Personalized Summary

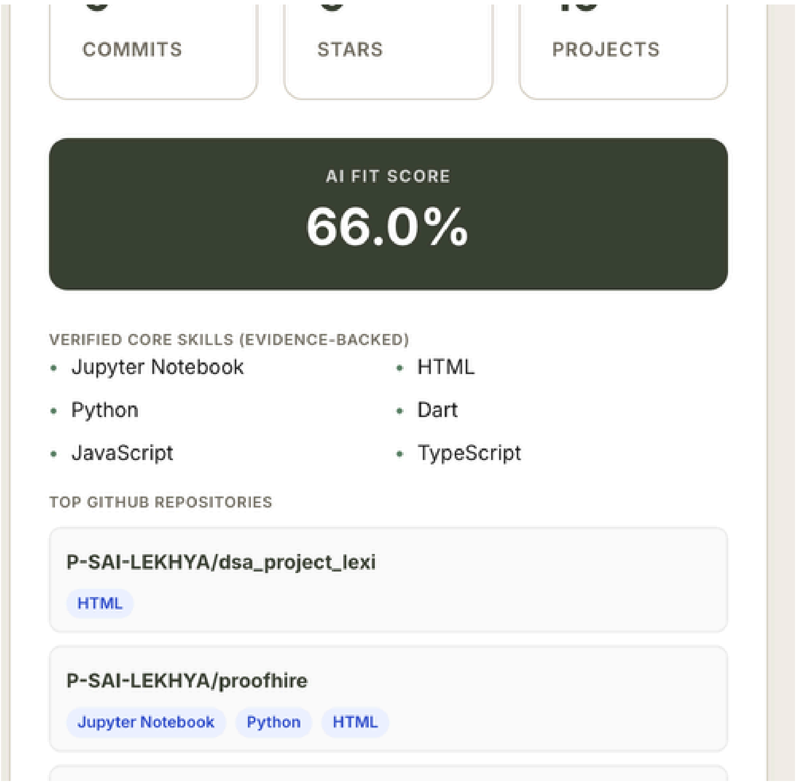


Resume is parsed

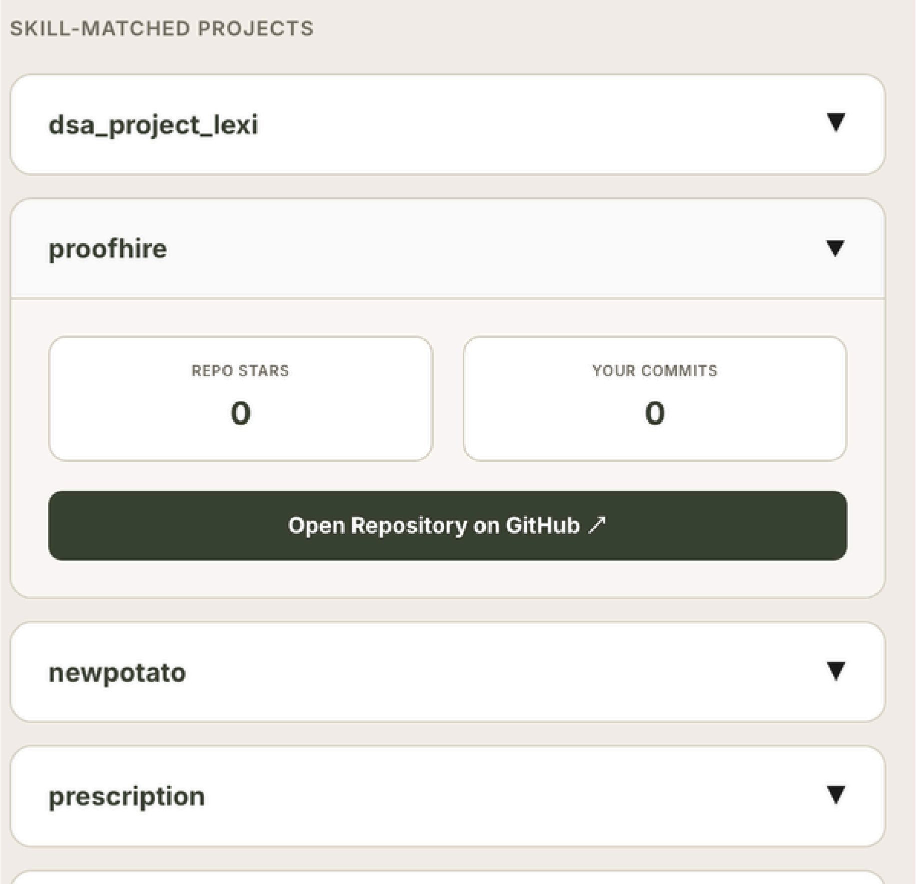


Generates knowledge graphs and skill analysis using resumes and GitHub repositories.

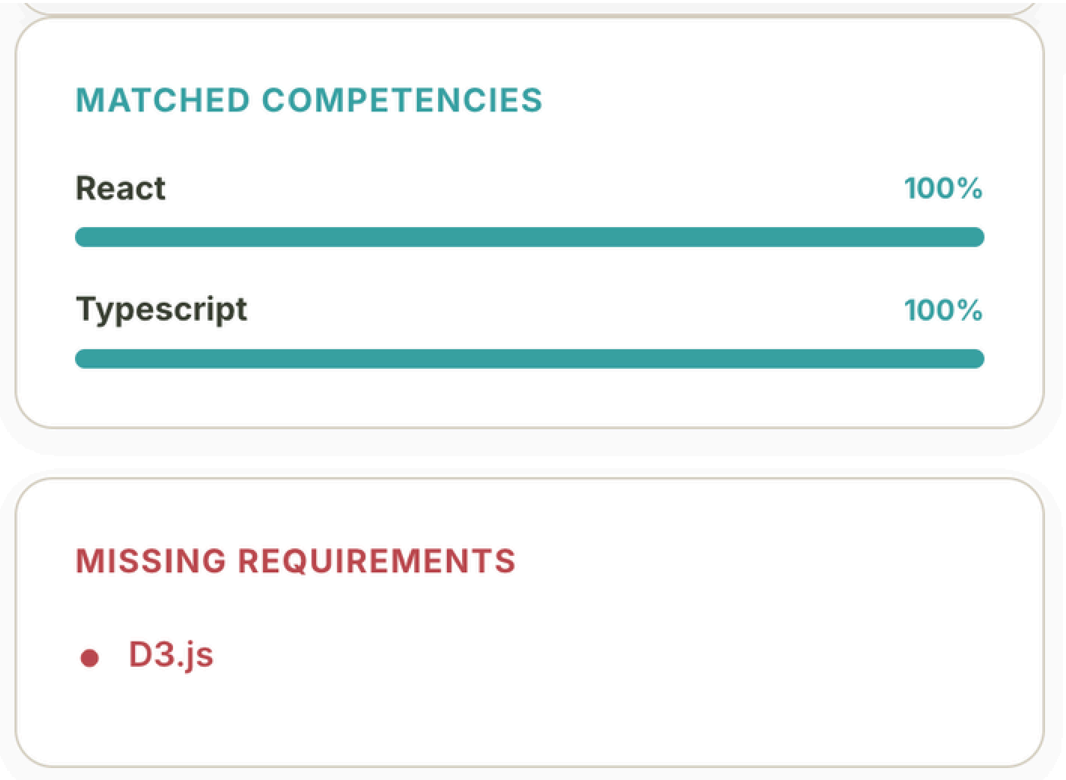
Proof-Hire : Multimodal Talent Intelligent Platform



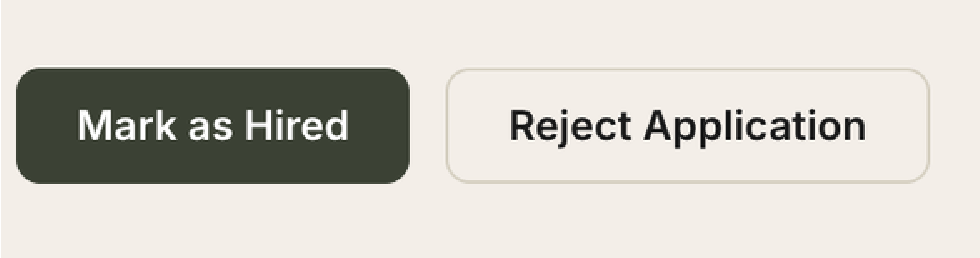
Github Scraping



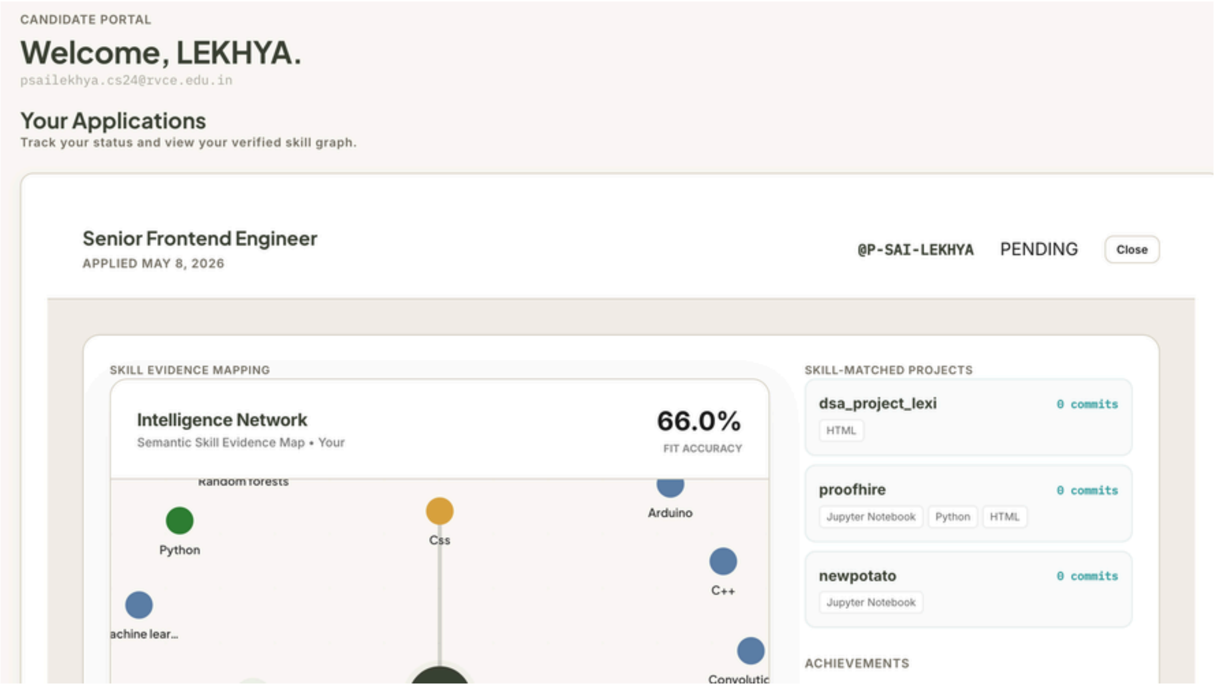
Skill Matched Projects



Matched and Unmatched skills



Based on all these, the recruite can hire or reject the candidate



Candidate can check his application status and his resume summary

A personalized roadmap is generated

Tech & Architecture

Backend Dependencies	
fastapi + uvicorn	REST API server
pdfplumber	Resume PDF parsing
requests	HTTP client for APIs
sentence-transformers	Semantic similarity embeddings (all-MiniLM-L6-v2 model)
scikit-learn	ML utilities (cosine similarity)
torch	Deep learning framework (transformers dependency)
networkx	Graph data structures for skill relationships

Backend Architecture

- Framework: FastAPI (production API server)
- Language: Python
- Server: Uvicorn

Frontend

- Framework: React 19.2.5
- Build Tool: Vite
- Language: JavaScript (ES modules)

Component	Input	Output
Frontend	User resume + job selection	HTTP POST to /api/apply
PDF Extractor	PDF bytes	Raw text
GROQ Parser	Resume text	Structured skills + metadata
GitHub Scraper	GitHub username	Repos, languages, tech stack
Semantic Engine	Candidate & job skills	Similarity scores, embeddings
Skill Graph Engine	Combined skills + job info	Graph structure, gaps

Tech & Architecture

AI usage(model, APIs, Framework)

Resume Parsing with GROQ LLM

- **Model:** openai/gpt-oss-120b
- Uses openai/gpt-oss-120b
- through the Groq API to extract skills, education, experience, and certifications from PDF resumes. Implemented in parse_resume_groq() with a regex-based fallback parser for reliability.

Semantic Skill Matching Engine

- **Model:** Sentence-Transformers (all-MiniLM-L6-v2)
- Built using Sentence Transformers (all-MiniLM-L6-v2) in semantic_engine.py. Converts skills into embeddings and computes cosine similarity to identify related technologies and semantic skill matches.

AI Candidate–Job Fit Analysis

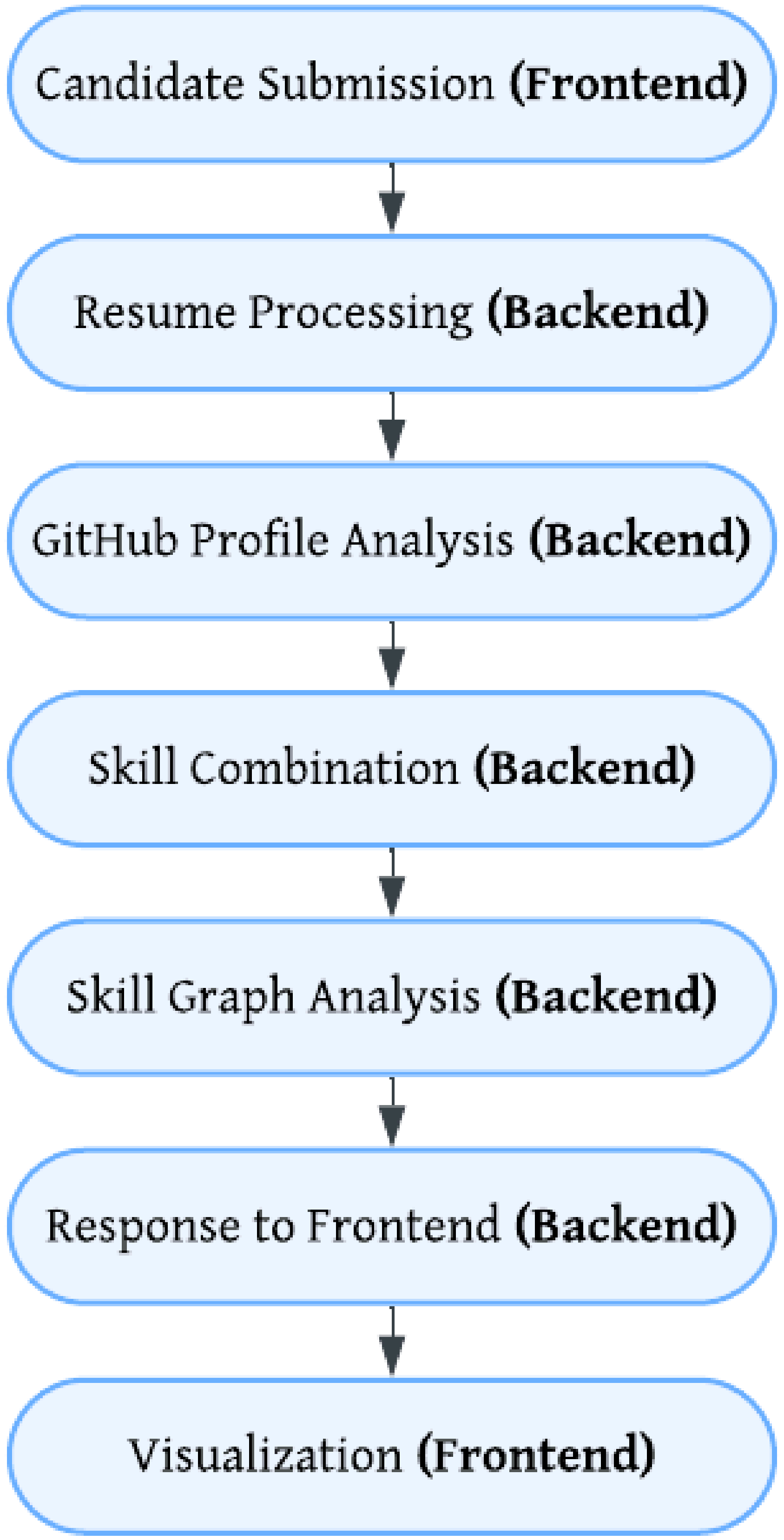
- **Model: openai/gpt-oss-120b**
- **(GROQ API)**
- Uses Llama 3.3 70B Versatile to evaluate candidate profiles against job descriptions. Generates AI-based fit summaries, identifies matching strengths, and highlights missing skills or hiring gaps.

GitHub Repository Analysis

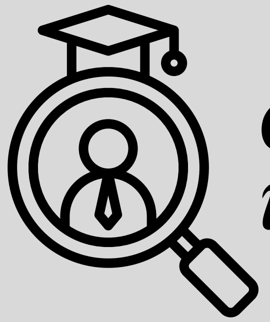
- Source: GitHub API
- Uses the GitHub API to analyze repositories, dependencies, and programming languages. Extracts technical evidence from files like package.json and requirements.txt to infer real developer skills.

Skill Graph Visualization

- Uses: NetworkX for graph structure, Streamlit/HTML for visualization
- Implemented in proofhire_skill_graph.py using NetworkX and D3.js. Generates interactive graphs showing relationships between candidate skills, job requirements, semantic matches, and GitHub-backed evidence.



Impact & Use cases



Campus Recruitment

Companies can quickly shortlist skilled students by analyzing resumes, GitHub projects, and technical capabilities instead of relying only on CGPA or keywords.

Startup Hiring

Startups with limited hiring teams can automate candidate screening and identify job-ready developers faster with proof-based evaluation.

Remote & Freelance Hiring

Recruiters can verify practical skills and project authenticity for remote candidates using GitHub activity and portfolio analysis.

Candidate Skill Development

Students and job seekers can identify missing skills, understand rejection reasons, and follow personalized learning roadmaps to improve employability.

Scale Potential

- Can scale across universities, startups, and enterprises
- Can expand beyond software hiring into design, research, and content domains
- Can integrate platforms like GitHub, Behance, Kaggle, and LinkedIn
- Useful for both recruitment and skill-development ecosystems
- Potential to become a universal proof-based talent intelligence platform

Differentiation

Proof-Based Skill Validation

Most hiring systems trust only resumes and keywords. Our platform verifies skills using real technical evidence from GitHub projects, portfolios, certifications, and coding activity, making candidate evaluation significantly more reliable and harder to manipulate.

Explainable AI Hiring Intelligence

Instead of giving a black-box match score, the system provides transparent reasoning through semantic skill graphs, gap analysis, and skill relationship mapping. Recruiters understand why a candidate fits a role, while candidates receive actionable improvement insights.

Verifies Skills Using Real Work

- Evaluates GitHub repositories, projects, and certifications instead of trusting resume claims alone.

Explainable AI Instead of Black-Box Scores

- Clearly explains why a candidate matches or lacks suitability for a role.

Understands Transferable Skills

- Uses semantic skill relationships to recognize related technologies and practical competency.

Reduces Resume Keyword Manipulation

- Prevents candidates from gaming ATS systems through simple keyword stuffing.

Supports Modern Career Paths

- Accurately evaluates freelancers, open-source contributors, and self-taught developers.

Provides Actionable Skill-Gap Insights

- Identifies missing skills and helps candidates improve for specific job roles.

Demo Link

GitHub Repo : https://github.com/Sm-LumeCode/Proof-Hire

Demo Link : https://drive.google.com/file/d/1lEcKgB3VABLlPd_JFdLSNXlnARfaLc5A/view

Document submitted : https://docs.google.com/document/d/1ErbSebWjmdCYwvQxr8BSYLtq1gdY_G53/edit